

Harbin Institute of Technology (Shenzhen)

Project Report

Image Processing (COMP5033)

2025–2026 Spring Semester

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1 Project Content

The goal is to find the boundary of objects in a binary image. The input is Figure_P2.jpg which is a 1024×1024 grayscale image. First I need to binarize it and then use morphological operations to get the boundary.

2 Method Description

Step 1: Binarization

I used Otsu's method to find the threshold automatically instead of guessing a value manually. The idea is to try every possible threshold from 0 to 255 and pick the one that best separates the image into two groups.

For each threshold t I calculate the between-class variance:

$$\text{variance} = w_0 * w_1 * (\text{mean}_0 - \text{mean}_1)^2$$

where w_0 and w_1 are the fractions of pixels in each group. The threshold with the highest variance is the best one. For this image it came out to $T=107$.

Step 2: Morphological Erosion

Erosion shrinks the white regions by removing the outermost pixels. For each pixel, I check its 3×3 neighbourhood – if all 9 pixels are white the pixel stays white, otherwise it becomes black.

To find the boundary I subtract the eroded image from the original:

$$\text{boundary} = \text{original} - \text{eroded}$$

This leaves only the pixels that were on the edge. I did this for both the white regions and the black regions (by inverting) and then combined them to get all edges.

3 Experiment Results and Analysis

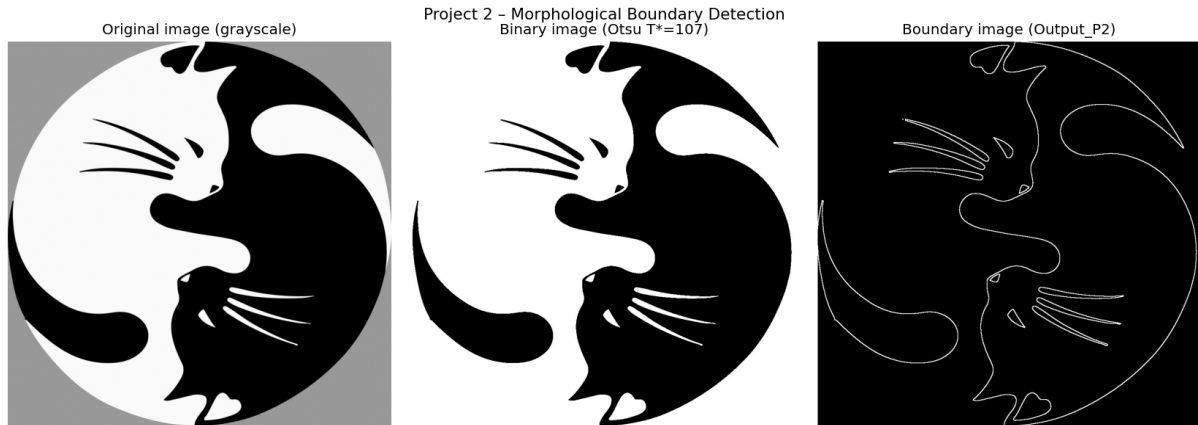


Figure 1. Original image, binary image after thresholding, and boundary output.

The threshold of 107 worked well – the binary image clearly separates the white cat from the black cat and the gray background. The gray border around the image became background which makes sense.

The boundary image shows a thin line around all the shapes. You can see the outer circle, the curve between the two cats, the whiskers and the face details. Total boundary pixels were 27,460.

One problem I noticed is that very thin lines like the whiskers partly disappear after erosion because the 3×3 window is too big for them.

4 Summary

Implementing Otsu's method from scratch was the hardest part. I had to keep track of the running mean and weight for each threshold which was a bit tricky to get right.

I learned that erosion is a simple but effective way to find boundaries – you just subtract what was eroded and you get the edge. I also learned that Otsu's method is really useful because you don't need to manually choose the threshold.